

ShelfScan: An End-to-End Embodied Framework for Automated Library Book Inventory, Recognition, and Sorting

Ling Ma*
Sun Yi*
maling920827@sjtu.edu.cn
yisun@sjtu.edu.cn
Shanghai Jiao Tong University
Shanghai, China

Cong Tian
Shanghai Jiao Tong University
Shanghai, China
tiancong123@sjtu.edu.cn

Wanru Yang
Shanghai Jiao Tong University
Shanghai, China
wryang@sjtu.edu.cn

Yahong Lu
Shanghai Jiao Tong University
Shanghai, China
luyahong@sjtu.edu.cn

Yue Ding
Shanghai Jiao Tong University
Shanghai, China
dingyue@sjtu.edu.cn

Feng Zhou
Shanghai Jiao Tong University
Shanghai, China
zhoufeng78@sjtu.edu.cn

Abstract

With library collections and circulation demands rapidly increasing, traditional manual inventory and shelving are no longer able to meet requirements for real-time, efficient, and precise management. To overcome the limitation of existing automation solutions that focus on isolated subtasks without an end-to-end loop, we introduce ShelfScan—an embodied intelligence system that integrates localization, RFID inventory, visual recognition, and robotic grasping into a single pipeline. By combining SLAM + iBeacon localization, RFID scanning, CRAFT+CRNN OCR with RAG-based vector retrieval, and one-shot demonstration grasping, ShelfScan automates the full “inventory→recognition→sorting” workflow. We validate the inventory and recognition modules in a real reading-room deployment and demonstrate over 95% grasp success in a high-fidelity simulation.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Robotic planning*; **Artificial intelligence**.

Keywords

ShelfScan; RFID Inventory; Retrieval; One-Shot Demonstration; Autonomous Book Sorting

ACM Reference Format:

Ling Ma, Sun Yi, Wanru Yang, Yue Ding, Cong Tian, Yahong Lu, and Feng Zhou. 2025. ShelfScan: An End-to-End Embodied Framework for Automated Library Book Inventory, Recognition, and Sorting. In *Proceedings of Companion of the 2025 ACM IJ/PUC (UbiComp Companion '25)*. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3714394.3756247>

*Both authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp Companion '25, Espoo, Finland

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1477-1/25/10
<https://doi.org/10.1145/3714394.3756247>

1 Introduction

Intelligent book management is the key driver for enhancing operational efficiency and user experience in modern libraries[6]. As collections and circulation volume surge, traditional shelf-by-shelf or trolley-based RFID inventory methods can no longer meet the demands for real-time accuracy and efficiency—they are time-consuming and slow to detect and correct mis-shelved books[7]. Existing automated and robotic solutions typically target only a single subtask (e.g., RFID scanning or mechanical retrieval) and lack a closed-loop pipeline from environment perception through mis-shelved detection to intelligent re-shelving[14].

To overcome these limitations, we propose ShelfScan—an end-to-end embodied intelligence[13] system that automates the full “Inventory → Recognition → Sorting” workflow in large-scale library environments. The main contributions of our work are summarized as follows:

- **End-to-End Closed-Loop Framework:** Seamlessly integrates SLAM/iBeacon localization, RFID scanning, OCR and vector retrieval, and robotic re-shelving into one pipeline.
- **OCR+RAG Precision Matching:** Embeds OCR’d spine text in a vector database for fast, accurate book ID and attribute lookup.
- **One-Shot Demonstration Strategy:** Derives reusable push – grasp – place trajectories from a single human demo, minimizing training overhead.

2 Related Word

2.1 Inventory and Localization

Traditional library inventory uses handheld RFID scanners or fixed gates, offering limited coverage and no real-time error detection on dense shelves. Systems like TagSLAM[10] and RF-SLAM[12] combine mobile SLAM with RFID for improved tag localization in warehouses and retail, but they aren’t designed for compact library stacks or mis-shelving detection and thus can’t be directly applied to automated book rearrangement.

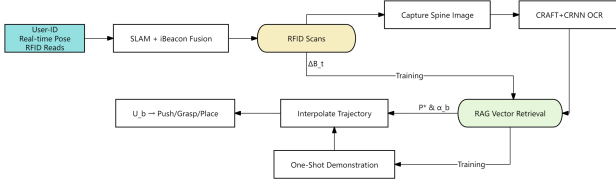


Figure 1: Model frame diagram

2.2 Spine Text Recognition

Classic spine-text OCR methods, for example CRAFT + CRNN[2], excel at reading narrow regions but struggle to match against vast catalogues. Retrieval-Augmented Generation[8] addresses this by embedding OCR outputs and querying a vector database to fetch exact titles and metadata. By tightly coupling OCR with vector retrieval, we achieve the precision required by libraries holding tens of thousands of volumes.

2.3 Grasp Execution

Traditional robotic grasping relies on extensive demonstrations or deep reinforcement learning, resulting in high training costs and slow deployment. One-shot methods such as YOTO[16] generate reusable trajectories from a single demonstration but have never been paired with RFID/OCR detection in libraries. Our system is the first to integrate mis-shelving detection with one-shot grasping to enable fully automated detect→retrieve→reshelve operations.

3 Model

This section provides an overview of the ShelfScan framework: the system uses BLE localization and RFID scanning to inventory the shelf and detect mis-shelved books. For each, it captures a spine image, uses OCR with RAG-based retrieval to match text and locate the book's slot, and retrieves its dimensions. Finally, it generates and executes push, grasp, and placement commands for the robotic arm(Fig. 1).

3.1 Inventory Module

In the ShelfScan system, the inventory module combines SLAM and iBeacon measurements to localize the robot and detect mis-shelved books in a unified process, with SLAM providing an initial pose ℓ_t^{SLAM} . We then refine this estimate by minimizing the RSSI error over N iBeacons (located at s_j with reference strength A_j and path-loss exponent n_j), yielding the fused pose

$$\delta \ell_t = \arg \min_{\delta \in \mathbb{R}^2} \sum_{j=1}^N \left(A_j - 10, n_j \log_{10} |\ell_t^{\text{SLAM}} + \delta - s_j|_2 - \widehat{\text{RSSI}}_j \right)^2. \quad (1)$$

The fused robot pose is calculated by adding the correction $\delta \ell_t$ to the initial SLAM pose ℓ_t^{SLAM} . Next, with $M = 4$ RFID antennas $\{a_1, a_2, a_3, a_4\}$, each antenna-tag reading (b_i, a_k) defines $\mathcal{R}_t = \{(b_i, a_k) \mid b_i \in \mathcal{B}, a_k \in \{a_1, \dots, a_4\}\}$. Using the fused pose ℓ_t and the antennas' calibration parameters, a mapping function $f_{\text{map}}(a_k, \ell_t)$ assigns each tag b_i to its true grid position on the shelf.

We then define the set of mis-shelved books as:

$$\Delta \mathcal{B}_t = \{b_i \in \mathcal{B} \mid (b_i, a_k) \in \mathcal{R}_t, f_{\text{map}}(a_k, \ell_t) \neq \text{ShelfPos}(b_i)\} \quad (2)$$

where $\Delta \mathcal{B}_t$ is the collection of books to be verified and re-shelved.

3.2 Recognition Module

The system first captures a frontal shelf image I_{cam} and uses the CRAFT text detector to identify candidate text regions $P = \{P_k\}$. Each region is affine-rectified and processed by a CRNN recognizer to extract text sequence t_k . These sequences are embedded into vectors $\mathbf{v}_{t_k} = \text{Embed}(t_k)$ and compared using cosine similarity with the embeddings of mis-shelved books $\{\mathbf{v}_b \mid b \in \Delta \mathcal{B}_t\}$ to find the best match. The best match is obtained by

$$b^*, P^* = \arg \max_{b \in \Delta \mathcal{B}_t, P_k \in \mathcal{P}} \cos(\mathbf{v}_{t_k}, \mathbf{v}_b) \quad (3)$$

where b^* identifies the specific book, and P^* its precise spine location in the image. Simultaneously, the physical attributes $\alpha_{b^*} = (w_{b^*}, h_{b^*}, d_{b^*})$ previously retrieved for b^* are returned to the manipulator module as input parameters for grasp planning.

3.3 Grasp Execution Module

Based on the localized spine region P^* , the identified book b^* , and its retrieved physical dimensions $\alpha_{b^*} = (w_{b^*}, h_{b^*}, d_{b^*})$, the system computes the three key waypoints for manipulation: the grasp point p_{b^*} , a lateral push target $(p_{b^*} + \Delta_{\text{right}})$, and the correct slot location $p_{\text{place}} = \text{ShelfPos}(b^*)$. It then generates the corresponding control sequence:

$$U_b = [\pi(q_{\text{home}}, p_b + \Delta_{\text{right}}), \pi(p_b + \Delta_{\text{right}}, p_b), \pi(p_b, p_{\text{place}})]. \quad (4)$$

where q_{home} is the robot's home configuration and $\pi(q_{\text{start}}, q_{\text{goal}})$ denotes the trajectory planner.

3.4 One-Shot Grasp Policy

The study employs a one-shot demonstration[5] approach structured into three stages: problem formulation, demonstration capture, and demonstration reuse.

Problem Formulation: We frame the shelf-retrieval task as a one-shot trajectory generation problem. The state space S includes the manipulator's joint configuration and grasp point p_b , while the action space A consists of continuous joint-space trajectories. The goal is to learn a mapping $\pi : (q_{\text{home}}, q_{\text{goal}}) \mapsto \tau$, where τ is a smooth trajectory from the start configuration q_{home} to any goal configuration Q_{goal} .

Demonstration Data Capture: The operator manually demonstrates keyframe actions from the initial pose q_{home} to the book reading pose $q_{\text{push}} = p_b + \Delta_{\text{right}}$, then to the grasping pose $q_{\text{grasp}} = p_b$, and finally to the placement pose $q_{\text{place}} = \text{ShelfPos}(b)$. These four keyframes are recorded in real-time, forming the teaching dataset.

One-Shot Trajectory Reuse: Using the recorded keyframes, Cartesian interpolation generates three trajectory segments—push, grasp, and place—without additional teaching. For each target book b , the process is as follows: 1. $\tau_{\text{push}} = \pi(q_{\text{home}}, q_{\text{push}})$ 2. $\tau_{\text{grasp}} = \pi(q_{\text{push}}, q_{\text{grasp}})$ 3. $\tau_{\text{place}} = \pi(q_{\text{grasp}}, q_{\text{place}})$

4 Experimental Design

To validate the effectiveness of the proposed ShelfScan system, this study evaluates its key modules using real-world library data and a simulation environment. The following sections briefly describe the dataset composition, simulation setup, evaluation metrics, and experimental results.

4.1 Dataset

The evaluation is conducted in a real-world reading-room section comprising 255 shelving units and a total of 53,557 volumes. We constructed three complementary datasets:

Book Vector Library: Metadata (ID, RFID, title, year, author, publisher) and location (x, y, z) were extracted for each book. Titles were embedded into 768-dimensional vectors using a multilingual Transformer model, paired with physical dimensions (w, h, d), creating a vector database of 53,557 books.

Field Inventory Test Set: The system was tested with all 53,557 books in the target reading room for one week using SLAM, iBeacon localization, and RFID scanning to detect misplaced books across 255 shelving units.

Image Set: 1,785 high-resolution shelf images were captured from 255 units. 1,250 images were used to train the CRAFT+CRNN text-detection models, and 535 images formed the test set. All images were manually annotated with spine-text bounding boxes to assess model accuracy.

4.2 Simulation Environment Construction

In the study, we used SketchUp[4] to recreate the reading-room shelving area at full scale, with each shelf measuring 90 cm $\times$ 28 cm $\times$ 25 cm and seven tiers. Books were modeled as rectangular prisms based on their dimensions (w, h, d) from the Book Vector Library. All 255 shelf units and 53,557 book models were imported into the Genesis simulation environment[1], positioned according to their (x, y, z) coordinates. The robot’s URDF was loaded, with a simulated front-view camera, RFID antenna plugins, and SLAM + iBeacon localization set to match the real setup for one-shot grasping experiments.(Fig. 2)

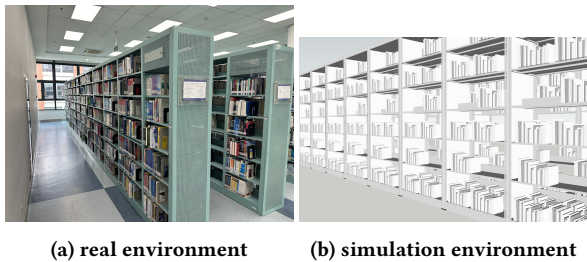


Figure 2: Book shelf: (1)real environment; (2)simulation environment

4.3 Baselines

We evaluate ShelfScan against strong baselines on two tasks. **Text Recognition:**(1) Tesseract OCR[11]. A general-purpose open-source

engine applied directly to spine crops. (2)Easy -OCR[9]. A PyTorch-based end-to-end OCR framework supporting multilingual scene text. **Shelf Manipulation:**(1) Action Chunking Transformers (ACT)[15]. Uses a transformer-based visual encoder on RGB observations. (2) Diffusion Policy (DP)[3]. Applies a diffusion model to point-cloud inputs with a PointNet++ encoder.

4.4 Metrics

Inventory: Over a one-week period, the system conducted daily full-shelf scans of 255 shelving units (53,557 books), for a total of 7 passes. Each book was scanned once per pass to compute the RFID read success rate and misplaced-book recall rate.

Recognition: Recognition Module: On the 535-image test set, we conducted five evaluation runs. In each run, the system processed all annotated spine regions and measured the character-level text recognition rate as well as the book-matching rate based on the retrieved titles.

Grasp: We selected 100 test books and performed 5 independent grasp trials per book (500 trials in total). For each trial, we recorded the completion of the push→grasp→place sequence to calculate the grasp success rate, and we logged the end-effector trajectory length (in meters).

4.5 Limitations

While the ShelfScan grasping strategy has been validated in simulation, several constraints remain. Experiments were limited to shelf slots where book thickness did not exceed two-thirds of the slot width, excluding dense configurations. Only books with spines fully facing outward and uniformly oriented were targeted, omitting inward- or downward-facing spines. Books were tightly packed with no intentional gaps, so the effect of variable spacing was not assessed. Additionally, the Genesis simulation does not account for real-world factors like friction, deformation, and dynamic inaccuracies in robot-shelf interactions.

5 Results

This section presents results by module: In the Inventory phase, we evaluate read success rate and misplaced-book recall at different scan speeds. In the Recognition phase, we report text detection, OCR accuracy, and book matching success using RAG-based retrieval, compared to baselines. In the Grasping phase, we test the one-shot policy’s grasp success rate, trajectory length, and execution time in simulation.

Inventory Results: During the one-week evaluation, scan speeds were increased from 2.5 cm/s to 10.0 cm/s. As speed rose, the average number of unread books per pass grew from 1.7 to 40.0, and the read success rate declined from 99.91% to 97.90% (Table 1). Additionally, out of 500 deliberately misplaced books, the system correctly detected 485, achieving a misplaced-book recall rate of 97% (Table 2)(Fig. 3).

Recognition Results: We evaluate our spine-text pipeline using three metrics: segmentation accuracy (75.6%), which measures the proportion of correctly identified text regions; OCR accuracy (72.4%), comparable to leading OCR baselines; and final accuracy (97.56%), achieved by integrating OCR with RAG-based

Table 1: Comparison of Test Results at Different Scanning Speeds

Run	Speed 2.5 cm/s		Speed 5.0 cm/s		Speed 7.5 cm/s		Speed 10.0 cm/s	
	Unread	Read Rate	Unread	Read Rate	Unread	Read Rate	Unread	Read Rate
1	118	99.78%	589	98.90%	1253	97.66%	1504	97.19%
3	0	100.00%	0	100.00%	107	99.80%	316	98.41%
5	54	99.90%	0	100.00%	213	99.59%	372	99.31%
7	43	99.92%	141	99.74%	643	98.80%	1601	97.01%
Avg.	48.20	99.91%	257.07	99.52%	455.23	99.15%	1520.86	97.16%

Table 2: Misplaced-Book Detection Recall and Counts Over Seven Consecutive Runs

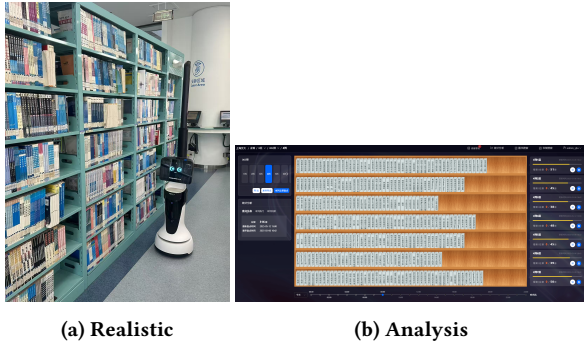
Run	Recall Rate(%)	Detected Books
1	95.60	478
3	97.2	486
5	98.20	491
7	95.80	479
Avg.	97.00	485

Table 3: Recognition Module Performance Comparison

	Seg Ac(%)	OCR Acc(%)	Final Acc(%)
Tesseract OCR	80.02	83.41	82.96
EasyOCR	90.08	88.67	90.35
ShelfScan(Only OCR)	92.84	87.69	91.25
ShelfScan	92.84	87.69	97.56

Table 4: book placement

	Book Extraction		Book Placement	
	Push	Grasp	Place	Secure
ACT	415/500	369/500	396/500	402/500
DP	435/500	374/500	463/500	459/500
ShelfScan(ours)	475/500	463/500	489/500	484/500

**Figure 3: ShelfScan Inventory: (a)Realistic, (b)Analysis****Figure 4: Example images of book recognition: OCR segmentation result.**

vector retrieval, outperforming OCR-only methods by over 5%.(Table 3)(Fig. 4).

Grasping Results: In our grasping evaluation, the task is divided into extraction (push and grasp) and placement. In the extraction stage, the manipulator completes 500 trials on 100 target books with a one-shot trajectory, achieving a 92.8% success rate, an average

trajectory length of 0.85 m, and an execution time of 3.2 s. Failures occur when a book's thickness exceeds the slot width or the spine is too tilted. In the placement stage, the arm places each book at the target slot's rightmost position, with a 97.2% success rate, a 1.8 cm average placement error, and an execution time of 2.6 s.(Table 4).

6 Conclusion

This work introduces ShelfScan, an end-to-end system that integrates localization, RFID inventory, visual recognition, and robotic manipulation to automate the "inventory → recognition → sorting" pipeline. ShelfScan has been successfully deployed in a real reading room, with its RFID and OCR+RAG modules validated on-site, and its grasping strategy demonstrated in a simulator. Future work will implement one-shot grasping on physical hardware and use reinforcement learning to enhance reliability and generalization in dynamic environments.

References

- [1] Genesis Authors. 2024. Genesis: A Generative and Universal Physics Engine for Robotics and Beyond. <https://github.com/Genesis-Embodied-AI/Genesis>
- [2] Youngmin Baek, Bado Lee, Dongyoon Han, Sangdo Yun, and Hwalsuk Lee. 2019. Character region awareness for text detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 9365–9374.
- [3] Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song. 2023. Diffusion policy: Visuomotor policy learning via action diffusion. *The International Journal of Robotics Research* (2023), 02783649241273668.

- [4] Aidan Chopra. 2012. *Introduction to google sketchup*. John Wiley & Sons.
- [5] Yan Duan, Marcin Andrychowicz, Bradly Stadie, OpenAI Jonathan Ho, Jonas Schneider, Ilya Sutskever, Pieter Abbeel, and Wojciech Zaremba. 2017. One-shot imitation learning. *Advances in neural information processing systems* 30 (2017).
- [6] Chandimal Jayawardena, Shyam Reyal, KR Kekirideniya, GHT Wijayawardhana, DGIU Rupasinghe, and SYRM Lakranda. 2021. Artificial intelligence based smart library management system. In *2021 6th IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE)*, Vol. 6. IEEE, 1–6.
- [7] Satish Kamble, Sachin Meshram, Rahul Thokal, and Roshan Gakre. 2014. Developing a multitasking shopping trolley based on RFID technology. *International Journal of Soft Computing and Engineering (IJSCE)* 3, 6 (2014), 2231–2307.
- [8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in neural information processing systems* 33 (2020), 9459–9474.
- [9] Binod Kumar Pattanayak, Anil Kumar Biswal, Suprava Ranjan Laha, Saumendra Pattnaik, Bibhuti Bhusan Dash, and Sudhansu Shekhar Patra. 2023. A novel technique for handwritten text recognition using easy OCR. In *2023 International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)*. IEEE, 1115–1119.
- [10] Bernd Pfrommer and Kostas Daniilidis. 2019. Tagslam: Robust slam with fiducial markers. *arXiv preprint arXiv:1910.00679* (2019).
- [11] Ray Smith. 2007. An overview of the Tesseract OCR engine. In *Ninth international conference on document analysis and recognition (ICDAR 2007)*, Vol. 2. IEEE, 629–633.
- [12] Chong Wu, Zeyu Gong, Bo Tao, Ke Tan, Zhenfeng Gu, and Zhou-Ping Yin. 2023. RF-SLAM: UHF-RFID based simultaneous tags mapping and robot localization algorithm for smart warehouse position service. *IEEE Transactions on Industrial Informatics* 19, 12 (2023), 11765–11775.
- [13] Fei Xia, Amir R Zamir, Zhiyang He, Alexander Sax, Jitendra Malik, and Silvio Savarese. 2018. Gibson env: Real-world perception for embodied agents. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 9068–9079.
- [14] Lu Yahong, Ma Ling, Yang Wanru, and Shi Xiaohua. 2023. Research on the practice and application of intelligent book inventory robot technology – Taking the library of Shanghai Jiaotong University as an example. *Journal of University Library* 41, 01 (2023), 44–51. doi:10.16603/j.issn1002-1027.2023.01.007
- [15] Tony Z Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. 2023. Learning fine-grained bimanual manipulation with low-cost hardware. *arXiv preprint arXiv:2304.13705* (2023).
- [16] Huayi Zhou, Ruixiang Wang, Yunxin Tai, Yueci Deng, Guiliang Liu, and Kui Jia. 2025. You Only Teach Once: Learn One-Shot Bimanual Robotic Manipulation from Video Demonstrations. *arXiv preprint arXiv:2501.14208* (2025).